

Exercise 1: Singleton Pattern

```
class Logger {
    private static Logger instance;

    private Logger() {
    }

    public static Logger getInstance() {
        if (instance == null) {
            instance = new Logger();
        }
        return instance;
    }

    public void log(String message) {
        System.out.println("Log: " + message);
    }
}

public class SingletonTest {
    public static void main(String[] args) {
        Logger logger1 = Logger.getInstance();
        Logger logger2 = Logger.getInstance();

        logger1.log("First Message");
        logger2.log("Second Message");

        if (logger1 == logger2) {
            System.out.println("Only one Logger instance exists.");
        } else {
            System.out.println("Multiple Logger instances exist.");
        }
    }
}
```

Exercise 2: Factory Method Pattern

```
interface Document {  
    void open();  
}
```

```
class WordDocument implements Document {  
    public void open() {  
        System.out.println("Opening Word Document");  
    }  
}
```

```
class PdfDocument implements Document {  
    public void open() {  
        System.out.println("Opening PDF Document");  
    }  
}
```

```
class ExcelDocument implements Document {  
    public void open() {  
        System.out.println("Opening Excel Document");  
    }  
}
```

```
abstract class DocumentFactory {  
    public abstract Document createDocument();  
}
```

```
class WordFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new WordDocument();  
    }  
}
```

```
}
```

```
class PdfFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new PdfDocument();  
    }  
}
```

```
class ExcelFactory extends DocumentFactory {  
    public Document createDocument() {  
        return new ExcelDocument();  
    }  
}
```

```
public class FactoryTest {  
    public static void main(String[] args) {  
        DocumentFactory wordFactory = new WordFactory();  
        Document wordDoc = wordFactory.createDocument();  
        wordDoc.open();  
  
        DocumentFactory pdfFactory = new PdfFactory();  
        Document pdfDoc = pdfFactory.createDocument();  
        pdfDoc.open();  
  
        DocumentFactory excelFactory = new ExcelFactory();  
        Document excelDoc = excelFactory.createDocument();  
        excelDoc.open();  
    }  
}
```